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Multimodal text-emoji fusion using deep neural networks for text-based emotion detection in online communication

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Abstract

The task of emotion detection in online social communication has been explored extensively. However, these studies solely focus on textual cues. Nowadays, emojis have become increasingly popular, serving as a visual means to express emotions and ideas succinctly. These emojis can be used supportively or contrastively, even sarcastically, adding complexity to emotional interpretation. Therefore, incorporating emoji analysis is crucial for accurately extracting insights from social media content to support decision-making. This paper aims to investigate to what extent the usage of emojis can contribute to the automated detection of emotions in text messages with a focus on online social communication. We propose an emoji-aware hybrid deep learning framework for multimodal emotion detection. The proposed framework leverages the feature-level fusion of textual and emoji representations, incorporating conventional and recurrent neural networks, to learn the fused modalities. The proposed approach was extensively evaluated on the GoEmotions dataset with different performance metrics. The experimental results indicate that emoji features can significantly improve emotion classification accuracy, highlighting their potential for enriching emotion understanding in online social communication.

Keywords: Multimodal, Fusion, Emoji, Text-based emotion detection, Deep learning, Early fusion

Introduction

The era of Web 4.0, a symbiotic or mobile web, symbolises the advancement of internet technologies to their next stage. This advancement will have a symbiotic relationship between humans and machines, where machines will learn from human interactions and help humans make decisions. The Internet has become an integral part of human life, where people generate social content by communicating through social media sites and reviews by sharing experiences and opinions on topics of interest. Likewise, with the advent of Artificial Intelligence (AI), people interact with intelligent dialogue machines in conversational AI systems. Understanding the sentiments and emotions behind this massive amount of online information has become a necessity. Emotional expressions and detection have become crucial parts of human–machine interaction [1] People

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express deep and complex emotions with a handful of words, and it has remained a challenge for machines to identify emotions and sentiments.

Human communication involves a variety of senses or modalities. These modalities work in concert to clarify concepts and emphasize ideas in discussions by resolving ambiguity. Similarly, nowadays, people are using various modalities in digital communications. People have adopted diversity in writing styles when it comes to communication with Individuals. One of the popular writing styles people favour these days is emojis in their interactions. The Oxford Dictionary states [2]," Emoji is a small digital image or icon used to express an idea or emotion in electronic communication". The word "emoji" is borrowed from the Japanese language, and it is partitioned as" e", meaning"picture", and"moji" meaning" letter or character". Emojis consist of smilies and hearts and are essentially the subsequent natural development of emoticons such as:) And:D. These days, emoticon means objects represented by standard character sequences, and emoji means self-reported labels as a piece of visual information users provide to express emotions in textual interactions to improve interpretability, particularly in short, complex interactions. Emojis can be found in various genres, such as smiles and people, animals and nature, food and drinks, travel and places, activities and objects. There are thousands upon thousands of emojis, each with a distinct meaning. This visual language is currently used as the de facto standard for online communication and can be found on Facebook, Instagram, WhatsApp, Twitter, and other major websites.

This visual information [2] in written text can express affective states, making it an appropriate means of expressing sentiment and emotion. These emojis and the text represent the user's emotional state more faithfully. Analysing the text without emojis would lead to the loss of important information that conveys people's feelings or clarifies the meaning of the text. In some instances, such as sarcastic phrases [3] a user may convey positive emotions in the text, but by utilising emoticons or emojis, they may convey negative emotions. In these situations, emojis might assist in determining the user's actual feelings. Emojis and emoticons are visual cues that aid in understanding natural language. For example, I'm satisfied with the service! 😡

In contrast, emojis are also not considered a direct expression of the emotions. Emojis can have several meanings based on the context and other emojis since they are defined without precise semantics [4]. So, text and emojis can play a complementary role in text-based emotion detection to comprehend people's emotions. Table 1 presents the sample text along with the accurate emotion label, the text emotion label, and the emoji emotion label. Despite their prevalence as language form, emojis and their underlying semantics have not been widely studied from a natural language processing (NLP) perspective. The complementary role of text and emojis is not explored in-depth by researchers who have combined features of text and emojis. So, in this work, the authors propose treating emojis as a new modality with text to enhance emotion detection. The text with emojis could accurately communicate the genuine feelings or emotions of the people they are trying to express.

The usage of deep learning techniques has increased with the introduction of big data [5]. Specifically, in sentiment analysis and emotion detection research, deep learning techniques have played significant roles. Deep learning techniques improve accuracy and performance by automatically learning and extracting information. Techniques like

Table 1 Sample Sentences with emotion label of text from the dataset, emotion label of emoji and valid sentence label

Sample Text	Emotion Label of Text	Emotion Label of the Emoji	True Emotion Label
I'm scared to even ask my mom I might get yelled at 😟	Fear	Sadness	Fear
Sorry [NAME]! 😗 😗 😗	Neutral	Admiration/Love	Sadness/ Disappointment
Omg [NAME] and his dad walking out together is so cute 😍	Neutral	Admiration	Admiration
He blocked me right after 😂	Neutral	Happy	Sadness
🤣 wow... I read it all in a serious tone until the end! 🤣	Excitement	Amusement	Amusement
Do you wash your hands with hand sanitizer? 🤔	Curiosity	Curiosity	Curiosity
To bad they didn't SHOOT the low life 😡 😡 😡 😡	Angry	Angry	Angry
That's not a denial 🤔	Disapproval	Nervousness	Approval

Convolutional neural network (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and their bi-directional variants, transformer-based models like BERT have proved highly effective in natural language processing tasks such as sentiment analysis, emotion detection. Advanced deep learning techniques [6] such as Generative Adversarial Net- works (GAN), Autoencoder (AE), Graph Neural Networks and Attention mechanisms are progressively used in NLP tasks.

Emojis embedded in the text can help to recognize feelings or emotions; this paper aims to identify emotions using deep learning techniques from text-based online communication with emojis. This study makes comprehensive use of emojis in text, as well as the relation between text and emojis. A multimodal deep learning model based on convolutional and recurrent networks with emojis is proposed for emotion detection in text-based online communication. Therefore, this multimodal model we develop must determine which underlying emotion class the input corresponds to given one or more input modalities. The authors used an emoji-text integrated GoEmotions dataset to propose a multimodal deep-learning model for better emotion detection. The main contributions of this paper are summarized as follows:

1. We perform a comprehensive literature review on multimodal studies in natural language processing applications. Similarly, we review sentiment and emotion classification of emojis-related studies in social media analysis.
2. We propose an emoji-ware multimodal hybrid framework, Multimodal Text- Emoji Fusion, based on a convolutional and recurrent neural network for emotion detection classification for text-based online communication.
3. We investigate the role of emojis in the feature-level fusion of text and emojis in emotion detection and multimodality in text-based emotion detection.
4. We conduct an intensive empirical analysis of the proposed model on the GoEmotions dataset using different performance measures.

The remainder of this paper is organized as follows. Sect. "Related Work" introduces the related work on multimodal fusion, unimodal and multimodal emotion detection, discussing deep learning-based techniques in emotion detection. Sect. "Multimodal Fusion" provides an overview of the dataset. A detailed explanation of the emoji-aware proposed framework is found in Sect. "Early Fusion". Sect. "Late Fusion" provides a detailed description of the experimental setup and the analysis of the results obtained from the GoEmotions dataset. Sect. "Intermediate Fusion" provides concluding remarks and presents the future research directions.

Related work

Emotion detection has become increasingly popular in the academic and research communities recently. There are various ways to detect human emotions, such as speech and facial emotion recognition. Similarly, through EEG signals, emotions can be detected. However, text-based emotion detection needs to be explored considering the latest developments in online text communication and the progress to increasing the emotional intelligence of human–computer interaction. As people are using multiple visual information such as emojis, emotion recognition from text and emojis needs to be studied. This section surveys literature from three perspectives: (1) Multimodal Fusion (2) Unimodal text-based emotion detection, (3) Multimodal Emotion detection using text and emojis.

Multimodal fusion

In today's online communication era, multiple modes are used to communicate and present ideas and information. Information is disseminated through written text, audio, and visuals, including images, videos, and emojis. Social networking sites are accountable for the continuous increase in the amount of information offered in visual forms. The potential for communication that visual representation offers is diverse. According to [14], the world as it is presented differs from the world as it is narrated or written. The term"multimodality" describes the presentation of information by combining or'in a sense fusing' many media formats. This fused representation of different media like text and visual or text and audio helps users to describe and understand the context in a profound manner. Multimodality in natural language processing tasks has remained in focus in recent years in the form of different modalities. Tasks include caption generation, speech recognition, sentiment analysis, and emotion recognition, where visual and audio with text modalities have been used. A summary of various representative multimodality-based NLP task works is provided in Table 2. The authors analyzed these works based on the modalities used, the kind of fusion, the approach employed, and the tasks for which it is utilized. Figure 1 shows the basic outline of the multimodal system. The process of integrating or fusing data from two or more modalities to make a prediction, classification or regression is known as fusion in multimodality. The multimodal data fusion process is characterized by the level at which the fusion of input modalities occurs in the network, like early fusion, late fusion, and intermediate fusion. Figure 2 presents the overview of the early and late fusion systems.

Table 2 Literature survey based on types of modalities used, fusion types used, and type task performed using multimodal fusion

References	Type of modalities	Fusion types	Technique	Task
[7]	Visual, text, acoustic	Early fusion/cross-modal	Deep Learning—Encoder with an attention mechanism	Sentiment analysis
[8]	Visual, text, acoustic	Early fusion	Deep learning—LSTM, BERT, CNN	Sentiment analysis
[9]	Text, acoustic	Late fusion	Deep Learning—Attention with auto- encoder, Bi-LSTM	Emotion recognition
[10]	Visual, text, emoji, acoustic	Early fusion	Deep Learning—Bi-GRU, Gated attention mechanism	Sarcasm detection
[11]	Image, text	Early fusion	Deep Learning—RNN, Pre-trained CNN	Machine translation
[12]	Text, emoji	Late fusion	Deep learning—Bi– GRU with attention	Emotion detection
[12]	Text, emoji	Late fusion	SVM, Deep learning—Bi-LSTM, CNN	Sentiment analysis
[13]	Text, acoustic	Late fusion	Deep learning—Bi-RNN, Self-Attention, BERT	Emotion recognition
[2]	Text, emoji	Early fusion	Deep Learning—FastText, Pre-trained FastText	Emotion classification

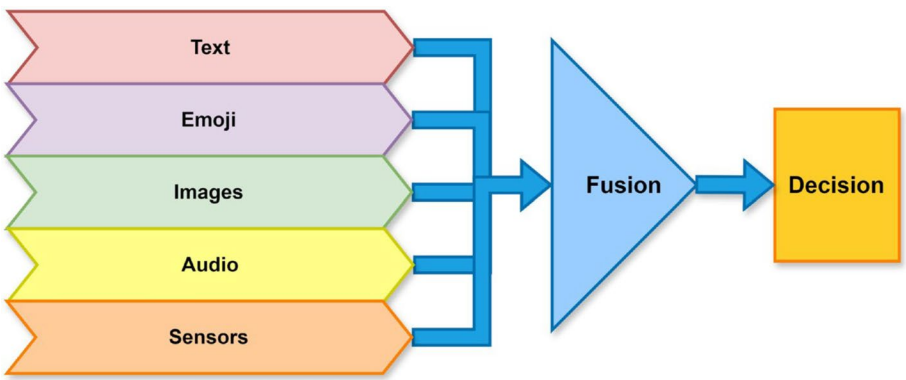


Fig. 1 Multi-modal Outline

Early fusion

It is the traditional way of fusion, also called a feature or data-level fusion. In early fusion, data fusing involves concatenating original data or extracted features, i.e., data transformed into features at the input level before input to a unified single model that accepts all modalities in one model. Data fusion takes place in many ways, typically pooling or concatenation. A Unified model can be any machine learning algorithm such as Random forest, SVM or deep learning model like CNN, RNN, etc. In early fusion, to accommodate multiple data modalities into a unified model, data transformation into a single feature vector is essential [15]. Proposed early data fusion by concatenating textual, acoustic and visual features in the multimodal stream for sentiment analysis.

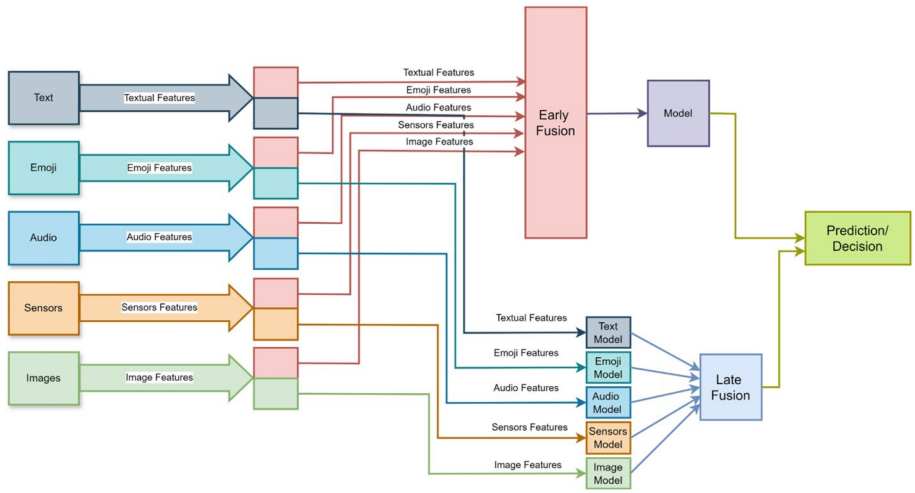


Fig. 2 Overview of early and late fusion multimodal system

Late fusion

Late fusion is also termed decision-level fusion as it aggregates predictions or results at the decision level. In late fusion, a number of algorithmic models are trained, typically one model per modality type. By combining the decisions from multiple models into a unified decision, a conclusion can be drawn. An ensemble approach inspires late fusion. In the ensemble approach, multiple models are trained on the same data, whereas in late fusion, multiple models are trained on different data. In late fusion, models must attempt to predict the same outcome. In late fusion, the number of models to be trained depends on the number of data modalities to be fused. To fuse the decisions from different models, different rules are used, such as the Bayes model, max-fusion, average fusion, etc. Late fusion is simpler than early fusion as compared from an architecture point of view. Data from different modalities are handled by multiple models individually and dealt with independently, so errors are also handled by models independently or uncorrelated, so late fusion gives good performance compared to early fusion [16]. Used an attention-based late fusion mechanism for task adaptation in natural language understanding. The authors developed a late fusion method capable of aggregating representation from intermediate layers of the pre-trained BERT model to adapt to a downstream NLU task.

Intermediate fusion

It is also called hybrid or joint fusion. In intermediate fusion, multiple models are trained but in a stepwise manner. The output of some models with input features from different modalities is given to subsequent models for outcome. Similar to early fusion, interaction effects between the parameters take place. Models are trained in a stepwise fashion so loss from the second model can be propagated back to the first model, updating weights for both models. Intermediate fusion is based on deep neural networks, allowing the most flexible fusion of features at different depths of model training. So, it is built in a very conscientious manner. Intermediate fusion provides significantly improved

performance compared to early fusion. In a deep learning multimodal context, a fusion of representations of different modalities into a single hidden layer, the model learns a joint representation of each modality. Different modalities can be fused simultaneously into a single shared representation layer or progressively using one or more modalities at a time [17]. Built a system that uses intermediate fusion to combine text, video, and audio modalities to detect aggression in surveillance systems.

Unimodal text-based emotion detection

A significant number of literature and research have been developed about text-based emotion detection. Emotion detection is an expanded version of sentiment analysis. In order to determine the sentiments of tweets, a BERT architecture incorporating CNN, RNN, and Bi-LSTM was assessed on the Twitter dataset [17]). The tweets were converted to vectors using Word2Vec. After evaluating various Word2Vec combinations with baseline models, BERT-Bi-LSTM showed the best results. [18]) curated multi-label emotion dataset from literature sources annotated for fine-grained categories. The study included thirty-eight emotion labels, with balanced samples in each category. The dataset was trained and assessed using the F1-Score using semi- supervised techniques like RoBERTa, BERT, and DistilBERT models. In [19], a hierarchical Bi-LSTM model with Glove embeddings was evaluated on the Twitter dataset. A Hierarchical Bi-CuDNNLSTM with NVIDIA CUDA deep neural network library used to detect emotions on emotiondatasetforNLP dataset. the authors also assessed the performance of Glove-CNN-LSTM, GloVe-Bi-LSTM, GloVe-CNN-Bi- LSTM and GloVe-CNN-Bi-GRU models. The transfer learning approach was used by [20] to capture the contextual meaning of the text and for a better understanding of the emotions. EmotionBERT model compared with baseline LSTM and Bi-GRU models using MELD dataset. While the aforementioned text emotion identification techniques improved performance, they exclusively used features found in text data. Emotional information is abundant in the visual content, such as emojis of the user data. So, when designing a precise text-based emotion detection system, it becomes necessary to incorporate text and emoji information to enhance emotion recognition performance by making it a multimodal system.

Multimodal emotion detection using text and emojis

When it comes to bringing more accuracy to text-based emotion detection tasks in online text communication, text and emojis work well together. Some researchers combined text and emoji modalities to improve the accuracy of text-based emotion detection. In one of the works on depression detection [21], multimodality of text and emojis was used. Benchmark dataset SentiEmoDD labelled with emotion, sentiment and depression symptoms used on a Multiview ensemble approach to detect the depression. The authors used BERT vectorization for generating text vectors and Emoji2Vec for emojis. This work used a late fusion of vectors. In machine learning, SVM linear kernel function, decision trees, Naïve Bayes, and KNN were used with a stacking approach. Similarly, in deep learning, CNN, RNN, LSTM, GRU models and stacking approaches were used. SVM outperformed the other individual approaches in the machine learning approach, but the stacking approach produced the best out- comes in the deep learning approach. The authors [12] created a Bi-LSTM model that combines emojis and text to

analyze emotions. the authors used the Sina social net- working site for Chinese microblog reviews. To obtain emoji embeddings, the emoji vectorization method was used in which the authors applied multidimensional emotion and emotional value to transform emojis into equivalent emoji vectors using word2vec. The text was also converted to vectors using word2vec. Linear SVM, random forest and LSTM models were evaluated on the dataset using macro precision, recall and F1-Score [12]. Presented a SEER model to capture the emotions of words and used emoticons to enhance the emotion detection task. The authors used the Bi-GRU model to capture semantic information, whereas statistical analysis of emoticons from datasets was used to extract emotional information from emoticons. Different machine learning and deep learning models, including SVM, GRU, LSTM and its bi-directional variants and CNN-Bi-LSTM, were assessed on NLPCC2013 and NLPCC2014 datasets using precision, recall and F1-Score. The authors [22] curated a large dataset of 58 k English Reddit comments with twenty-eight emotion classes. It is an emoji-integrated text dataset annotated using principal preserved component analysis (PCA). The authors used transfer learning based on the BERT model and evaluated the F1- score, precision, and recall along with Bi-LSTM as the baseline model [23]. Presents emojis that could contribute to automating sentiment analysis. The authors created an Arabic-language Twitter tweet dataset. In order to integrate text and emojis at the feature, score, decision, and hybrid levels, the authors suggested early and late fusion. For text feature extraction, TF-IDF, CBOW and Skip-gram approaches were employed, whereas emoji frequency, lexicon-based, emoji CBOW and emoji skip-gram methods were utilized for emoji extraction. These features were classified using SVC and LR and evaluated using precision, recall, F1-score and accuracy [24]. Developed a speech act classification to determine the intent of Twitter. The authors generated a dataset from the speech act on Twitter and named it EmoTA. The dyadic attention mechanism was introduced to fuse the modalities in the intra-model and inter-model attention ways. The authors used BERT for text feature extraction and emoji2Vec representation for emoji feature extraction. Bi-directional LSTM was used in a multi-tasking framework using a fully shared and shared private model. When this multitask, framework was evaluated based on accuracy and F1-score, the multimodal results outperformed the unimodal ones.

Dataset

In order to train the predictive models in supervised machine learning algorithms, labelled datasets are needed. A significant amount of human effort is needed to cresate annotated or labelled datasets. One of the significant challenges in multimodality is choosing a multimodal dataset. The authors surveyed three datasets suitable for multimodal analysis with text and emojis. The authors studied GoEmotions [22], EmoContext [25], and TweetEval [26] datasets curated from social media sites such as Twitter or Reddit. After thoroughly looking at the datasets based on the number of records, number of emotions, and sources from where they were curated, the authors preferred to go with the GoEmotions. The GoEmotions dataset offers the most comprehensive coverage of expressed emotions and minimum overlap between emotions, so the authors preferred it. It was built on 27 emotion labels with neutral categories. These 28 emotion labels from the dataset are easily mapped further to the Ekman model of emotions. This

collection of 58 K records has been compiled from Reddit comments. It is available for public use*. Data was gathered from the platform from Reddit's founding in 2005 until January 2019.

The dataset contains the records with only text and emojis. Some records include simple text, text with one emoji, and text with more than two emojis. Although data contains two modalities, text and emojis, the dataset is annotated only for text information. It is observed from Fig. 3 that there is a significant discrepancy in terms of emotion frequencies. For example, adoration is 30 times more frequent than grief. Figure 3 presents the emotion categories with the number of samples. we have analyzed the total number of entries in the training, validation, and testing sets that include emojis incorporated in the textual description. It was observed that around 3.51% of the samples had emojis.

Proposed methodology

This section outlines the overall architecture for the multimodal framework using text and emojis and implementation details. Figure 4 displays the overall architectural diagram of the suggested system. Our approach is based on extracting features from two distinct modalities: text and emojis. They are described as follows:

Unimodal feature extraction

Text modality

We have developed two different systems for two different modalities. Both systems consist of two subparts. First, preprocessing, feature extraction, then combining features and model classifier to predict the emotion.

1. Text Preprocessing—The data must be pre-processed in order to extract and identify relevant features. Preprocessing is necessary since the content from the user is mainly composed of short text, slang terms, incomplete words, etc. Text pre- processing is crucial in preparing natural language data for effective analysis and model-

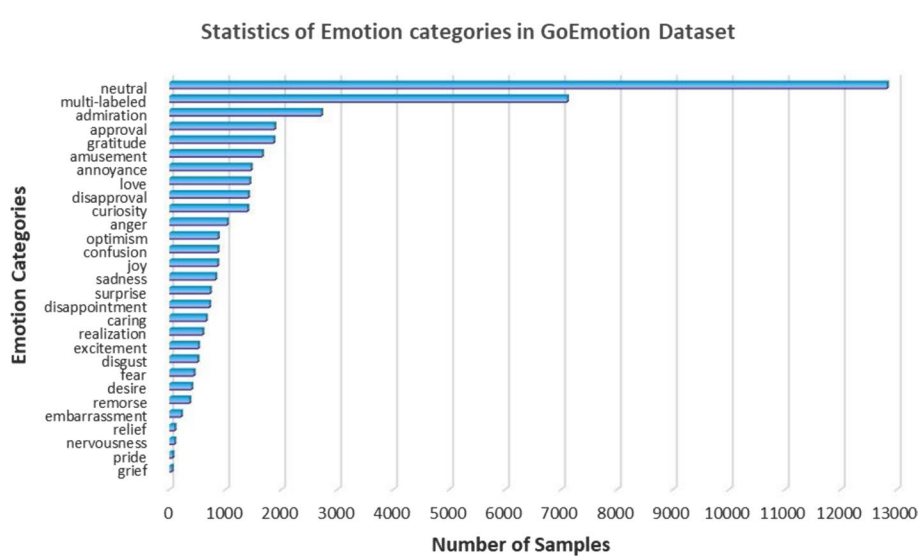


Fig. 3 Statistics on the total number of samples with single and multiple labels for each emotion category

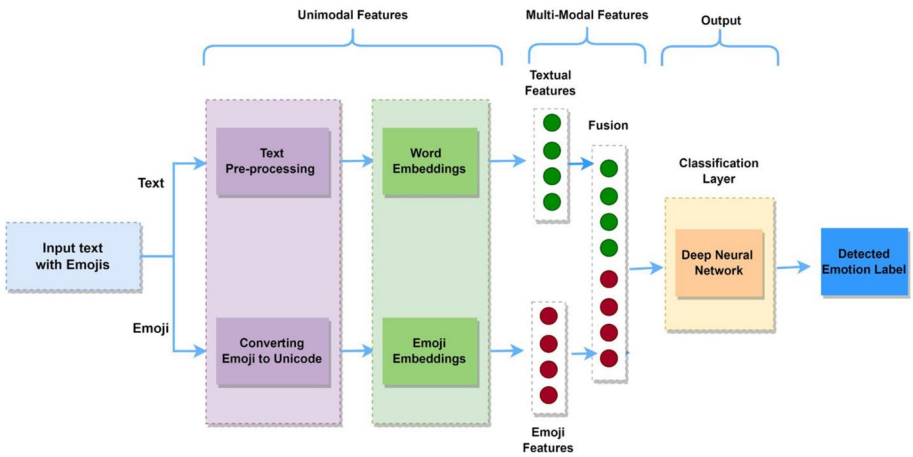


Fig. 4 Overall architectural diagram of the proposed network

ling. Textual data is pre-processed with data cleaning, normalization and vectorization.

(a) Data Cleaning: We have performed the following stages to remove noise and inconsistencies from the text data, typically including:

- Lowercasing: Converting all text to lowercase eradicates capitalization-based variations.
- Punctuation and digit removal: Omitting punctuation and numerical characters improves the focus on linguistic features.
- Stop word removal: Eliminating common words with minimal semantic content (e.g.,"the,""a," and"is") reduces noise and dimensionality.
- Hashtag and HTML tag removal: Removing non-linguistic elements like hashtags and HTML tags enhances textual clarity.

(b) Tokenization/Segmentation: In this step, the text is divided into distinct chunks known as tokens, which are usually words or morphemes. As a result, the original text is represented by a series of tokens that may be further processed and analyzed.

(c) Normalization: This step improves consistency and facilitates analysis by reducing morphological variants of words to their regular forms. Stemming and lemmatization are two typical methods of normalizing. We have applied lemmatization to identify the grammatical root of a word (lemma) based on its context and linguistics.

2. Textual Feature Extraction—Then, the second step is to extract the features from pre-processed textual data. Our approach leverages pre-trained NNLM embeddings to represent words as vectors within a d-dimensional space. These embeddings, trained on the GoEmotion corpus, provide a sophisticated comprehension of word

meanings and emotions. The NNLM algorithm [27],) introduced the idea of training word embeddings in a statistical neural network language model (NNLM). This novel methodology aimed to predict the subsequent word in a sequence by leveraging the previous words, effectively capturing the semantic and syntactic connections among them. NNLM embeddings, context-based representations, are token-based text embeddings trained on diverse English Google News 200B corpus. In this, each word (token) is mapped to a 128-dimension fixed-length vector representing its meaning and relationships with other words, which means capturing semantic and syntactic relationships between words, even for out-of-vocabulary (OOV) words.

Extracting textual features from the vocabulary of NNLM as V, with cardinality $|V|$. It involves the following steps:

- The first step is One-Hot Encoding, where each word from preprocessed input $w \in V$ is mapped to a one-hot vector of dimension $|V|$ and given by equation 1:

$$h_w = (h_{w1}, h_{w2}, ..., h_w|\mathrm{V}|) \tag{1}$$

where:

h_{wi}: 1 if and only if $i = w$ and 0 otherwise

These vectors are sparse and computationally expensive due to their high dimensionality.

- The Second step, the embedding layer, is a linear transformation represented by a matrix $W \in R_{(d*|V|)}$, where d is the desired embedding dimension (here 128). The word embedding for $w \in V$ is then obtained as: $e_w = W * \mathrm{h}_w \in R_d$ This transformation projects the high-dimensional one-hot vectors into a lower-dimensional space while preserving semantic information. In the learning word embedding, the embedding layer parameters are learned during the NNLM training process. An objective function is used to optimize the prediction of surrounding words based on the current word embedding.
- In the final feature vector construction, word embeddings are combined into sentence embedding using the sqrt combiner.

Emoji modality

1. Emoji Processing—We have used emoji, a Python-based library for separating the visual representation of an emoji, to extract the emoji features from the input text. There are 1816 distinct kinds of emojis accessible in all.
2. Emoji Feature Extraction—Then, we employed Emoji2Vec, a pre-trained word embedding model specifically designed for emojis for emoji representation. This

model is based on a method designed to generate representations compatible with word2vec, a popular text-based word embedding framework. In this model, Emojis are mapped into a d-dimensional vector space according to their semantic similarity, as determined by the Unicode standard definitions of each one. So, embedding emojis into a vector space enables the model to comprehend emojis not as isolated symbols but as elements with semantic meanings, reducing the ambiguity to some extent. If an input text contains n_v emojis, we obtain the final emoji vector representation V as a matrix of dimensions $n_v * 300$, where each row represents the vector of an individual emoji. The final emoji vector representation V is obtained as a matrix $V \in_{(nv * dv)}$, capturing the semantic information of all emojis present.

Fusion of modalities

Early fusion or feature fusion approach is applied to combine the features from text and emojis. This approach merges information from different modalities by directly concatenating their extracted features. This generates an augmented feature vector in a higher-dimensional space, encompassing the combined information. The extracted feature vectors or embeddings of text modality and emoji modality are concatenated. It employs a simple feature-level fusion approach by concatenating the extracted textual and emoji features. Mathematically, let $\Gamma = (f_1, f_2, \ldots, f_d)$ be the textual feature vector with d dimensions, represented in R_d where $d = 128$, $E = (e1, e2, \ldots, en)$ be the emoji feature vector with n dimensions, represented in R_n where $n = 300$. The combined feature vector C is then obtained by directly concatenating these two vectors:

$$C = \left(f_1, f_2, ..., f_d, e_1, e_2, ..., e_n\right).$$

Model architecture

The combined feature vector C is made to pass through the modality encoder. The proposed multimodal emotion detection using a hybrid convolutional-recurrent model is shown in Fig. 5. In Fig. 5, red circles show the emoji representations, while green circles show textual representations. Similarly, the circle shape represents the recurrent component, and the square shape represents the convolutional component in the modality encoder of the system. The modality encoder is a neural network which predicts the final class label using the SoftMax function, as it is a multiclass classification detection problem. The Convolutional neural network layers are used for local feature extraction to identify the patterns that might be indicative of specific emotions and, similarly, recurrent layers to capture long-range dependencies and con- textual information from the sequence of text with emojis. Given an input text t with emojis, the proposed multimodal model calculates a probability for each 28-emotion label using a series of layers. These layers utilize the pre-trained word embeddings, where NNLM pre-trained embeddings

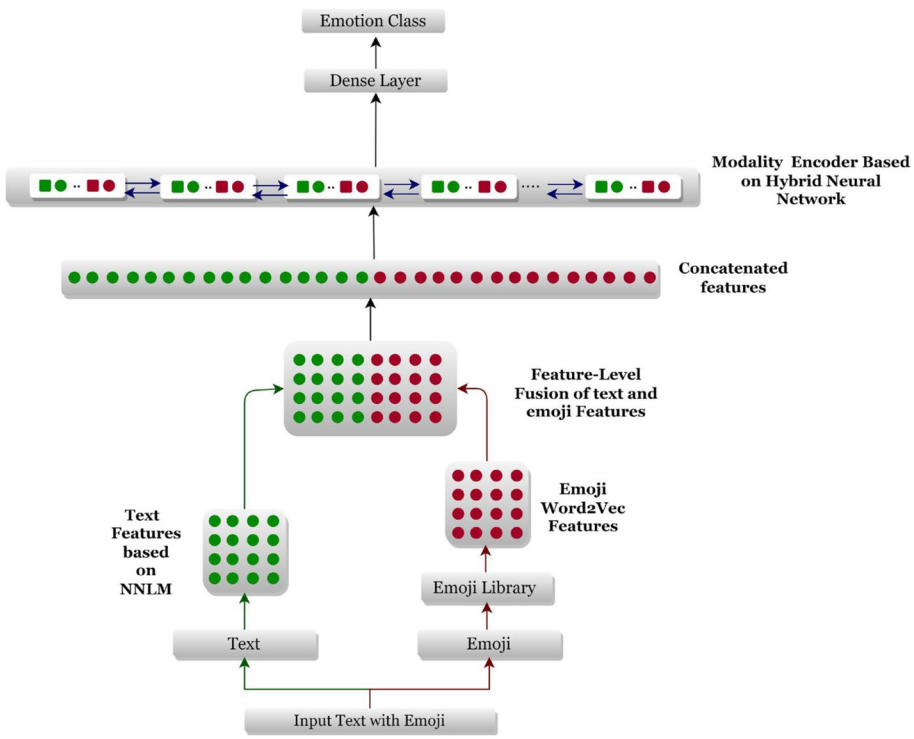


Fig. 5 Detailed architectural diagram of the proposed network

are used to convert each word in the phrase into a high-dimensional vector representation, and Emoji2Vec pre-trained embeddings are used in the case of emojis. Combined vectors capture semantic and contextual information about the word and emojis, allowing the model to learn essential features from the entire sequence. The authors experimented with the combined architecture of CNN- LSTM and CNN-Bi-LSTM. Similarly, the authors have used LSTM and bidirectional LSTMs (Bi-LSTMs). We adopted LSTM to understand the long-term dependencies with context and a Bi-LSTM network to capture the sequence's forward and backward context. LSTM, CNN-LSTM, BiLSTM, and CNN-BiLSTM models are proficient at catching sequential and contextual dependencies in data. So, when emojis and text are combined in an early approach, these models learn combined embeddings, allowing them to comprehend emoji meanings based on the surrounding textual context. Also, bidirectional models can process input from both directions, forward and back- ward, aiding the model in contextualizing emojis more effectively in broader sentences. CNN can capture local patterns and relationships and inherently shift-invariants. So, combining the characteristics of convolutional features with sequential modeling will enhance the extraction of local patterns, which may disambiguate emoji meanings to improve understanding of multimodal emotion detection.

1. Convolutional component: In the convolutional-recurrent multimodal model, initially, concatenated features (c_i) will be passed through convolutional layers. The one-dimensional convolutional layer is used where k filters are applied to the input text t

with emojis. Individually filter is represented as Γ of size s represented as a matrix of dimension $s * d$. Each filter slides across the sequence of s consecutive embeddings $_{(i:i+s-1)}$. This convolution operation produces a new feature c_i for each window and is expressed by Eq. 2

$$c_i = f\left(F * e_{(i:i+s-1)} + b\right) \tag{2}$$

where:

f: Non-linear ReLU function.

b: Bias-vector.

Each filter, in the convolution operation, slides across all possible windows of features in the sequence. This process generates a feature map, denoted as $cm = [cm_1, cm_2, \ldots, cm_{(n-s+1)}]$, where n is the sentence length and s is the filter size. In the first stage, if k filters are used, this operation creates k distinct feature maps. In this specific case, the convolutional layer employs 128 filters, resulting in the generation of 128 distinct feature maps. After the convolutional layer, a max pooling layer is applied to get 128 feature values corresponding to 128 feature maps. Then, these feature values are provided to the recurrent layer sequentially.

2. Recurrent component: The convolution layer outputs a series of feature vectors.

cm_1, cm_2, ..., cm_k, which are sequentially sent to the recurrent layer. For each combined vector (denoted by cm_i), LSTM computes only using one hidden state. At the same time, the Bi-LSTM computes it's forward and backward hidden states, denoted by h_{i-1} and h_{i+1}, respectively. These are obtained using the following equations Equation 3 and 4:

Forward hidden state:

$$h_{if} = LSTM_{fd}(cm_i, h_{i-1}) \tag{3}$$

Backward hidden state:

$$h_{ib} = LSTM_{bd}(cm_i, h_{i+1}) \tag{4}$$

The final hidden state matrix for the feature vector representation of user input, h_u, is then constructed by concatenating the forward and backward hidden states for each word in the sequence and is expressed by Eq. 5:

Final hidden state matrix:

$$H_u = [h_1, h_2, \ldots .h_n] \in R_{(n\ *2*dl)} \tag{5}$$

Here, n_c is the sequence length, dl is the number of hidden units in each LSTM layer, and h_u is the concatenation of h_{if} and h_{ib}. This final matrix, H_u, captures the

contextualized representation of the entire user input, serving as the input for subsequent processing. At each time step t, the recurrent network takes the input cm_t and the output of the previous step h_{t-1}. It then creates the current output h_t by applying a linear transformation following a non-linear activation function and expressed by Eqs. 6 and 7.

$$h_t = tanh(W_{cmh} * cm_t + W_{hh} * h_{t-1} + b) \tag{6}$$

$$y_t = tanh\left(W_{hy} * h_t + b_y\right) \tag{7}$$

where:

h_t: hidden state in time t.

cm_t:input feature vector from the convolutional layer in time t.

y_t: output in time t.

W_{cmh}:Weight matrix for input to the hidden connection.

W_{hh}:Weight matrix for hidden-to-hidden connection.

W_{hy}:Weight matrix for hidden-to-output connection.

b: Bias vector for the hidden layer.

b_y: Bias vector for the output layer.

The final classification stage takes the output from the recurrent layer as input. In simpler terms, the SoftMax function takes the recurrent layer's output (Eq. 8)(possibly containing high-dimensional representations) and transforms it into a probability distribution over the N possible output labels (emotions) for a given text input. And output layer is defined as follows –

$$P(e) = Softmax(y_t) \tag{8}$$

where:

$P(e)$: the probability distribution over emotion classes.

Experiments

Evaluation measures

In order to evaluate proposed multimodal system, we adopted accuracy, weighted precision, weighted recall, and weighted F1 score. Evaluating the performance of emotion detection models presents a challenge as an intrinsic imbalance in emotion class distributions within most real-world benchmark datasets. Accuracy alone is a biased measure in imbalanced classification problems. No single metric can definitively capture the model's effectiveness in such scenarios. Therefore, to gain a more comprehensive understanding of the model's performance, we have also used a classification report and confusion matrix to analyze the emotion class-wise performance as it is a multiclass classification problem. Similarly, accuracy and loss curves against time were used to show the performance of the proposed system. Distinct colors are utilized to depict various models in curves, while the continuous line signifies the accuracy and loss during training. On the other hand, dashed lines indicate the accuracy and loss of the validation.

Algorithm 1 Algorithm

Require:
Input: Input Text (t)
Output: Probability Distribution of Emotion Label
Ensure: Pre-processing
Text Lowercasing: Convert text (t) into lowercase letters
$t_{lower} = lowercase(t)$
Text Cleaning: Remove punctuation, special characters, emojis, digits, stop word, tags from lowercased text.
$t_{clean} = clean(t_{lower})$
Tokenization: Split the cleaned text into individual words
$(w_1, w_2, \quad , w_n) = tokenixe(t_{clean})$
Lemmatization: Normalize the words to their base form.
$(w_1, w_2, ..., w_n) = lemmatize(w_1, w_2, \quad , w_n)$
Word Embeddings: Convert each word (w) to a numerical representation using pre-trained NNLM word embedding having dimensions of 128.
$(f_1, f_2, ..., f_d) = Convert\ to\ word\ embedding\ (w_1, w_2, \quad , w_n, d)$

Emoji Preprocessing
Extracting Emoji from text: Extract emoji (e) from text (t)
$e_{extract} = extract(e, t)$
Emoji Embeddings: Convert emoji (e) to a numerical representation using pre-trained Emoji2Vec word embeddings(d=300)
$(e_1, e_2, ..., e_n) = ConvertoEmojiembeddings(w_1, w_2, ..., w_n, d)$

Fusion
Fusing features from both modalities: Concatenate the text embeddings and emojis embeddings to generate augmented feature vector.
$C = (f_1, f_2, ...f_d, e_1, e_2, ...e_n)$

For 50 **epoch**
Model Architecture:
Convolutional Layer: Extract the local feature from the sequence of word embeddings (e)
$c(i) = f\ (\sum(w_i^e * \boldsymbol{\Gamma}) + b)$
Max Pooling Layer: Reduces the dimensionality of the feature maps by selecting the maximum value.
$c_m = max(c(1), c(2), \quad , c(n - s + 1))$
Recurrent Layer: Captures the long-range dependencies between the words in the sequence.
$h_t = \tanh(W_{cmh} * cm_t + W_{hh} * h_{t-1} + b)$
$y_t = \tanh(W_{hy} * h_t + b_y)$
Dropout Layer: Drops a certain percentage of neurons during training to prevent over-fitting.
Fully-connected Layers: Fully-connected 64 layers with ReLU activation function.
$f(y_t) = max(0, y_t)$
SoftMax Layer: To convert the output of last fully-connected layer x into a probability distribution over 28 emotion classes.
$P\ (e)_i = softmax(x_i) = \frac{e^{x_i}}{\sum e^{x_j}}\ for\ all j(1. \ 28)$
Training: Monitor the training and validation loss and accuracy to avoid over-fitting using categorical cross-entropy loss function.
$L, P\ (e) = -\sum(y_i * \log P\ (e)_i)$

Experimental settings

The GoEmotion dataset was divided randomly into two parts of 80% 20% split for train and test sets, respectively. And again, from the training set, 20% was used as a validation set. The statistics of the training, validation, and test set are shown in Table 3. For single modality and multimodal classification, LSTM, Bi-LSTM and Hybrid architectures of CNN-LSTM and CNN-Bi-LSTM were adopted. To optimize performance, the

Table 3 Summary of hyperparameters values used in the proposed hybrid model

Hyperparameters	principal values
Convolutional parameters	
Embeddings	NNLM embeddings, Emoji2Vec embeddings
Filter size	128
Kernel size	5
Pooling	Max = 2
Stride	Default = 1
Padding	Same
Activation function	ReLU
Recurrent parameters	
Hidden units	64
Number of layers	1
Epochs	20
Activation functions	SoftMax
Loss function	Categorical Cross Entropy
Optimizer	Adam optimizer
Learning rate	0.001
Batch size	512

parameters were carefully selected through a what-if analysis. In a hybrid architecture, a one-dimensional convolution layer with 128 filters was used. This convolutional layer of kernel size five was used with the ReLU function. A max- pooling layer of size two was used in the subsequent layer. To encode fused modalities, LSTM with 64 hidden memory cells were used, followed by a flattened layer. So, Bi-LSTM versions resulted in a 128-dimensional representation vector due to the bidirectional nature of the layers. A fully connected layer of dimensions 28 was used in the subsequent layer. For model optimization, the Adam optimizer is employed. The Adam optimizer is used in this specific scenario, with a learning rate of 0.001 and a batch size of 512. Similarly, Sparse Categorical cross-entropy loss was used with a learning rate of 0.001 was used in the final experimental setting. The tests are performed on a computer that is equipped with a 12th generation Intel Core i7 CPU operating on the Linux subsystem, running at a frequency of 2.10 GHz, with 16 GB of RAM. The Jupyter framework is used for coding and running Python. The implementation is carried out via the TensorFlow Keras library.

Results and discussions

A series of experiments were conducted to evaluate the proposed multimodal approach. The proposed approach requires thorough analysis owing to the class imbalance present in the dataset, whereby the"neutral" class exhibits a much higher number of samples in comparison to the other classes. Imbalanced datasets might lead to biased predictions in the model, favouring the dominant class and disregarding the minority classes as unimportant. Therefore, it is essential to deal with this matter. In order to assess the model's effectiveness in these circumstances, we use measures like as precision, recall, and F1-score, which are particularly suitable for datasets with uneven distribution. An alternative method involves resampling the dataset by either under- sampling or oversampling. Oversampling mitigates the problem of class imbalance by augmenting the

quantity of data instances belonging to underrepresented classes. It does this by iteratively and randomly sampling data points from the minority class. We have implemented the Random oversampling approach.

For the baseline models, experiments were conducted in different categories. Experiments were conducted for only text modality, only emoji modality and combined modality (Text + Emoji) in varying combinations of the network architectures. As stated earlier, the goal of this study is to analyse the role of emojis in emotion detection; thus, the authors do not put focus on improving or analyzing the performance of emotion detection and treat it as an auxiliary task aiding the primary task. We show the comparison between our proposed approach and baselines, i.e., Baseline-1 [22], Baseline-2 [28] and baseline3 [29], which also made use of the same dataset. These baseline models are trained only on the textual features. This study distinguishes itself within existing research on this dataset by incorporating both textual and emoji features in its proposed approach. No prior work has explored the combined influence of these modalities on the task. Results are shown in Table 4. It summarizes the results of only text modality, only emoji modality and combined Text + Emoji modality with LSTM, Bi-LSTM, CNN-LSTM, and CNN-Bi-LSTM models in terms of various performance metrics. Accuracy curves and loss curves are also one of the evaluation measures. Distinct colors are utilized to depict various models in curves, while the continuous line signifies the accuracy and loss during training. On the other hand, dashed lines indicate the accuracy and loss of the validation in Fig. 6.

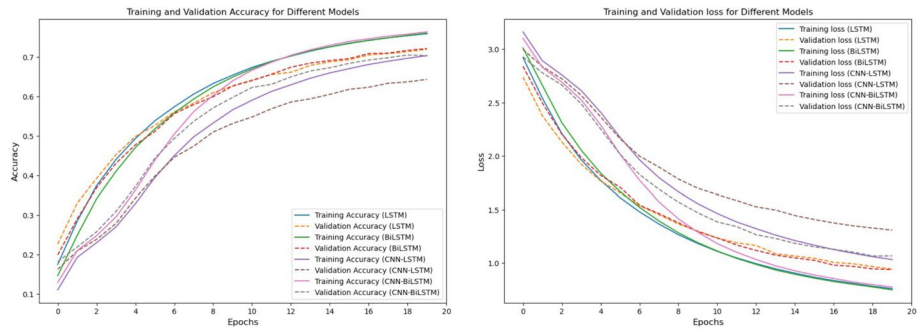
As visible in Table 5, the only textual modality provides the best results compared to only the emoji modality on the GoEmotions dataset. T and E represent textual features and emoji features, respectively. For only text modality setup, if we compare our results, there is a performance improvement of 23.0 points ↑ with Baseline-1, 3.0 points ↑ with baseline-2 and 6.4 points ↑ with baseline-3 in the F1-score. Table 5 shows the comparison of our best results of unimodal and multimodal with SOTA methods. For Text + Emoji modalities, we observe that multimodal setups outperform both unimodal modalities. We can observe that performance has remarkably improved with combining Text + Emoji modality and the highest overall performance given by the CNN-Bi-LSTM model with training accuracy of 77.80%, validation accuracy of 73.2% and F1-score of 71.68% with a multimodal approach. Figure 7 presents a comparison of models in terms of different performance measures.

Table 4 also displays the precision, recall, and F1-score metrics for the multimodal model. An investigation of precision and recall is essential for a thorough evaluation of a model's performance, especially when dealing with unbalanced datasets. Precision is the value obtained by dividing the number of correctly predicted positive outcomes by the total number of expected positive results. Recall measures the proportion of accurately identified positive outcomes out of all the actual positive instances in the data. All models have shown the good precision and recall values for combined modalities as compared to unimodal. It means models are good at predicting positive instances of labels when text and emojis are combined.

A confusion matrix is an effective tool for evaluating the efficiency of a model in multiclass classification tasks. This is a square matrix, which has an equal number of rows and columns. Each row and column in the matrix represents a distinct class. Each column

Table 4 Results of proposed multimodal models in terms of accuracy, precision, recall and weighted F1-score

Model	Only text					Only emoji					Combined (text + emoji)				
	Accuracy		Precision recall F1-score			Accuracy		Precision recall F1-score			Accuracy		Precision recall F1-score		
	Training	validation				Training	validation				Training	validation			
LSTM	60.82	57.34	53.81	57.57	54.47	43.81	35.91	36.63	35.91	32.31	75.94	72.01	69.91	72.24	70.64
CNN-LSTM	67.32	62.00	60.83	63.84	61.73	51.93	37.72	46.68	37.72	38.94	72.45	67.70	65.32	67.60	65.96
Bi-LSTM	69.64	61.80	68.37	70.67	68.80	50.37	36.95	42.18	36.95	35.63	77.03	72.15	69.61	71.97	70.08
CNN-Bi-LSTM	70.28	65.57	68.57	70.62	69.14	53.70	37.99	45.52	36.43	36.96	77.80	73.29	71.01	72.96	71.68



(a) Accuracy Curves (b) Loss Curves

Fig. 6 Accuracy curves and loss curves for proposed multimodal models on the GoEmotions dataset

Table 5 Comparison of SOTA methods with our results on the Empathetic dialogue dataset for emotion detection

No	Reference	Deep learning method	Performance metrics
1	Baseline—Text [22]	BERT	46.0
2	Baselines—Text [28]	RoBERTa	66.1
3	Baseline—Text [29]	GNN	62.7
4	Our best results—Text	CNN-Bi-LSTM	69.1
5	Our best results—Text + Emoji	CNN-Bi-LSTM	71.7

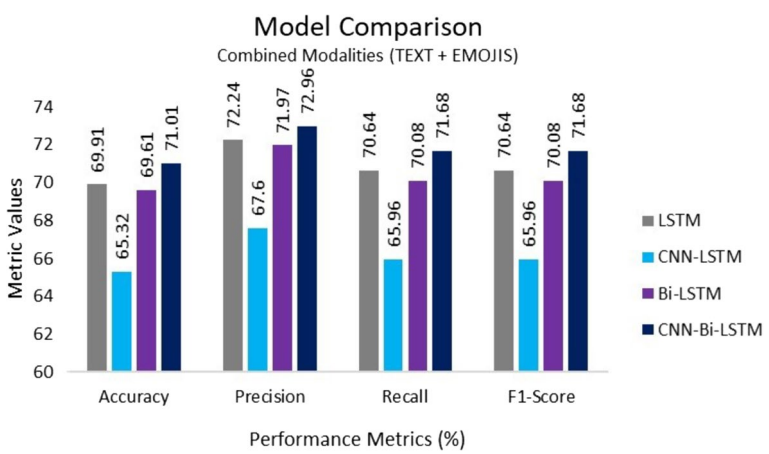


Fig. 7 Comparison of models for different performance measures

in the matrix represents the frequency of predictions made by the model for a certain class label, compared to the actual class label in the data. Optimally, the diagonal elements, which indicate correctly categorized instances, possess greater values, indicating successful predictions. Whereas off-diagonal elements shows misclassifications. In Fig. 8, all models shown diagonally high values which means correct predictions of the models in each emotion class. As the authors can figure out from confusion matrix of CNN-Bi-LSTM model has shown, “grief”, “relief”, “pride”, “nervousness” and “fear” are

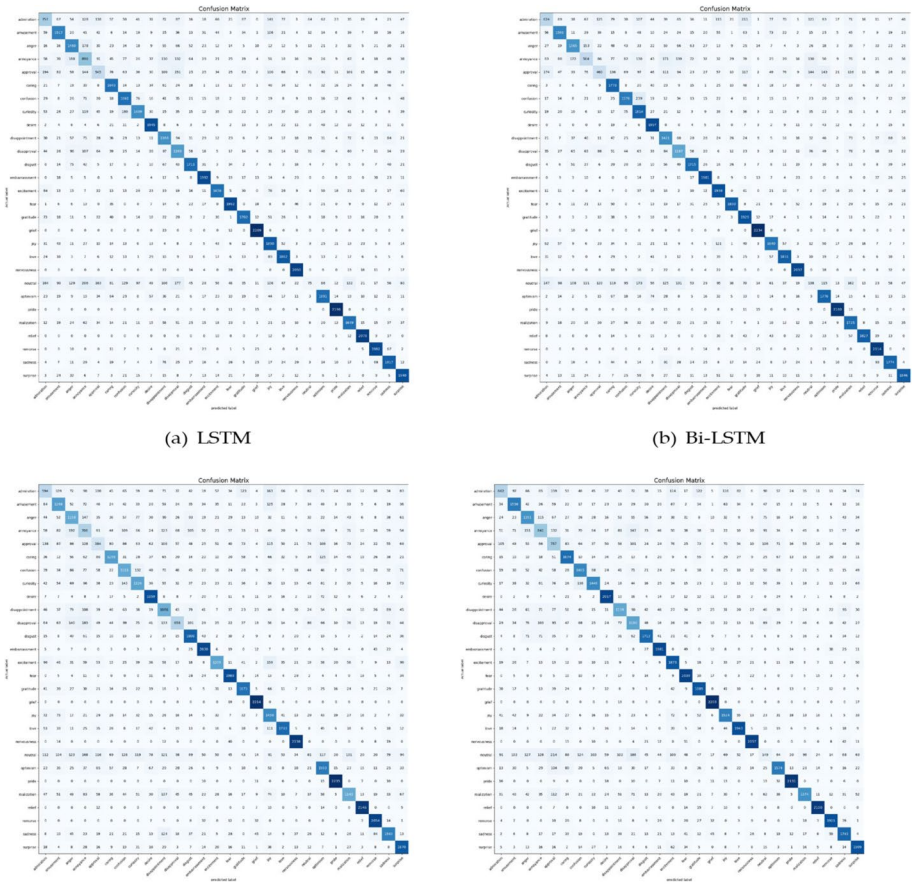


Fig. 8 Presents a confusion matrix for (**a**) LSTM, (**b**) Bi-LSTM, (**c**) CNN-LSTM, and (**d**) CNN-Bi-LSTM models

the emotions that are better predicted. Similarly, class "neutral" has the less number of correct classifications. There are some emotion class pairs where we could see misclassifications, like in between "neutral and approval", "neutral" and disapproval". The less accuracy for these emotions might indicates a similarity in the words, phrases and emojis used to convey them.

Ablation study

An ablation study is an effective technique used in artificial intelligence to analyse the specific impacts of various components of a model on its overall performance. It entails methodically eliminating or altering components of the model and analyzing the impact of these modifications on its output. To understand the effect of emoji and the effect of the convolutional component on the proposed model, we perform an ablation study. The results are shown in Table 5. For the setups, we observe that the best-proposed model outperformed w/o emoji and the proposed w/o convolutional component.

Table 6 Ablation study

Best setup	Accuracy in %	F1 score in %
Model w/o Text	37.99	36.96
Model w/o Emoji	65.57	69.14
Model w/o convolutional	72.15	70.08
Proposed Model	73.29	71.68

Table 7 Sample text with its corresponding predicted labels

Examples with text + emoji	Proposed systems Predicted label	Correct Prediction

# Example text input 	
user_text = "I am feeling happy 😊"
	1/1 [==========] - 1s 937ms/step
	Predicted emotion: relief
		Yes

# Example text input 	
user_text = "Feeling excited for the weekend! 🎉"
	1/1 [==========] - 0s 63ms/step
	Predicted emotion: excitement
		Yes

# Example text input 	
user_text = "I can't believe I have to work late again tonight! 😩💪 #grateful"
	1/1 [==========] - 0s 31ms/step
	Predicted emotion: excitement
		No

# Example text input 	
user_text = "I can't believe my car broke down on the way to the concert! 🙏 #blessed"
	1/1 [==========] - 1s 965ms/step
	Predicted emotion: gratitude
		No

# Example text input 	
user_text = "sorry dear! 😘😘😘"
	1/1 [==========] - 0s 31ms/step
	Predicted emotion: disappointment
		Yes

# Example text input 	
user_text = "I'm scared to even ask my mom ,I might get yelled at 😨"
	1/1 [==========] - 0s 31ms/step
	Predicted emotion: surprise
		No

# Example text input 	
user_text = "I am happy with the service! 😡😡"
	1/1 [==========] - 0s 31ms/step
	Predicted emotion: annoyance
		Yes

# Example text input 	
user_text = "Wow, that's rough. Wishing you the best! 😁"
	1/1 [==========] - 0s 32ms/step
	Predicted emotion: gratitude
		Yes

# Example text input 	
user_text = "but again, she/he will keep doing this 😊"
	1/1 [==========] - 0s 18ms/step
	Predicted emotion: realization
		Yes

Impact of emojis

Although our main focus is to study the effect of emojis on emotion detection, from Tables 4, 6 and 7, We observe that emojis are helpful in improving the performance of emotion detection. Empirically, we have shown that emoji helps emotion detection (Table 6). We took some examples from the dataset (Table 7) and off the dataset also, which are sentences that include emojis to show the effect of emojis. Each example has an implicit/explicit emotion with emojis. We have included examples in both ways where emojis are supporting text and where emojis are contrasting the text, such as sarcastically or ironically. Table 7 shows the sample text with an emoji with the predicted label. Table 7 shows the predicted label for examples and whether the proposed model predicted the label correctly or not by human evaluation. Interestingly, the predictions made by the model were correct in many sarcastic cases, but in some cases, they were incorrect. We observe that emoji's emotion plays an important role in emotion detection. In the example,"I am happy with service! 😡", the emotion displayed by the emojis is negative, whereas the text is positive. This aids the model in understanding the contradistinction between the emotion displayed by the text and the emoji. Thus, it correctly

interprets that the example is sarcastic. In the example,"I'm scared to even ask my mom; I might get yelled at 🤔", the model fails to comprehend the precise distinction between the text as the emojis do not provide any further distinguishable information to the model about it in training.

Real-world scenarios of proposed system

The proposed model has the potential to be used in real-world scenarios as it combines convolutional and recurrent layers with Emoji2Vec and NNLM (Neural Network Language Model) embeddings. Combining the strengths of the proposed model with domain-specific embeddings applies to a wide range of real-world scenarios, including natural language text and emojis.

- Integration into current frameworks—[30]

The proposed system can be integrated into existing systems or current frameworks where text and emoji-based emotion detection can be used as a supplemen-tary task. In chatbots or virtual assistants, the proposed model can be used to detect the emotions of input text and to generate responses empathetically. Emotions such as happiness, confusion, and frustration can be easily recognized by combining text and emoji cues, and more effective communication can be achieved. Similarly, in recommendation systems, Community Management Tools, and Customer-Centric Marketing Systems, the proposed model can provide more nuanced emotion detection by understanding the emotional tone of emojis with text.

- Domain-Specific Customization—[31, 32]

Various fields, including healthcare, e-commerce, and social media, have distinct usage patterns for emojis. In healthcare, emojis are used to denote symptoms or mental state through emotions. In e-commerce, emojis are used to indicate product characteristics or feelings related to product or service use. The proposed model can be trained using domain-specific datasets, allowing predictions to be customized to the distinct needs of each domain.

- Cross-Linguistic and Cultural Adaptability—[33]

Despite their widespread use, emojis may have variable meanings depending on the languages and cultures. As per cultures, people use different emojis to convey their thoughts and emotions. E.g., An emoji 🙏, for instance, may mean "thank you" in one culture and "prayer" in another. The proposed model, which utilizes Emoji2Vec embeddings to transform emojis into semantic vector spaces depending on their usage in natural language, enhances its ability to recognize intended meanings across many cultural contexts. Similarly, Cross-linguistic and cultural adaptability can addressed by training and fine-tuning the proposed model using multilingual datasets. To ensure it understands cultural and linguistic fineness in emoji use, the proposed model can be pre-trained on datasets annotated with different languages and then fine-tuned using domain-specific data, which can further enhance the interpretation of emojis.

Conclusion

This study presents a comprehensive review of work related to emoji-based emotion and sentiment analysis in online social communication. This work investigates the role of emojis in emotion detection and multimodality in text-based emotion detection. The authors explored the potential of multimodal text-emoji fusion for text-based emotion detection on the GoEmotions dataset using baseline deep neural networks. In the proposed hybrid framework of convolutional and recurrent networks, the authors effectively combined textual and emoji features, achieving superior performance compared to models relying solely on text. It highlights the significant emotional information embedded within emojis and their cooperative effect with textual cues. Additionally, the ablation study provided valuable insights into the contribution of each modality, revealing the importance of both textual and emoji features for optimal performance. So, the simplicity of the concatenation of two modalities at the feature level makes it a straightforward to implement and computationally efficient system. Additionally, combining features from various modalities has been empirically shown to improve prediction performance, leading to its widespread adoption within the research com- munity. Future research directions include investigating the impact of different emoji embedding techniques such as transformer-based embeddings can offer robust solutions for disambiguating emojis and improving the precision of the system to further enhance emotion detection accuracy in online social communication. Similarly, the late fusion approach of modalities could be evaluated to investigate the impact of emojis.

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Author contributions
Conceptualization, S.K., S.P., and K.K.; Methodology, S.K., S.P., and K.K.; Software, S.K.; Validation, S.K., S.P., and K.K., formal analysis, S.K., and S.P.; investigation, S.K., S.P., and K.K; resources, S.K. and S.P.; data curation, S.K., and S.P.; writing—original draft preparation, S.K.; writing—review and editing, S.P.; visualization, S.K.; supervision, K.K.; project administration, S.P.; funding acquisition, K.K. All authors have read and agreed to the published version of the manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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